



SQL BUSINESS ANALYSIS

Sales Data Sample

A wholesale distributor dataset, 2003–2005 — data quality audit, cleaning, and three business question pages, built end to end in PostgreSQL.

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APPROACH

Eleven checks, before a single business question

If the data can't be trusted, neither can anything built on top of it. Every check below was run before any revenue, product, or customer analysis began.

- ✓ **Structural overview** 2,823 rows, 25 columns, all data types correctly assigned
- ✓ **Duplicate records** Zero found — composite key on order number + line number holds cleanly
- ✓ **Cardinality** 307 orders, 92 customers, 109 products, 7 product lines, 19 countries — all reasonable
- ✓ **Category consistency** Status, product line, deal size, country, territory — no typos or casing issues
- ✓ **Numeric ranges** Quantity 6–97, sales \$482–\$14,082, MSRP \$33–\$214 — no impossible values

ONE CRITICAL DEFECT FOUND

PRICEEACH was stuck at a flat placeholder value of 100.00 in 1,304 of 2,823 rows — 46% of the entire dataset. Confirmed present in the raw source file, not an import artifact. Cross-referencing the same product code against MSRP and SALES showed genuine varied prices sitting right alongside repeated flat values on unrelated orders — not how real pricing behaves. This could not be used at face value for pricing analysis.

Three fields had missing data — each one investigated, not just flagged

ADDRESSLINE2

2,521 / 2,823 (89%)

No action needed

Population varies by country in line with real address complexity (Ireland 100%, Australia 79%, USA 10%) — field is only filled when a genuine second line exists.

STATE

1,819 / 2,823 (64%)

No action needed

Structurally expected — only USA, Australia, Canada, Japan use this field, and all four are 100% populated. UK shows minor capture inconsistency, not corruption.

POSTALCODE

76 / 2,823 (2.7%)

Left null, documented

Genuine gap, not structural. Traced to exactly 3 California-based customers. City/state present for all affected rows, so regional analysis is unaffected.

What changed, and what was deliberately left alone

Issue	Action Taken
Missing ADDRESSLINE2, STATE, POSTALCODE	Left as NULL — root cause already established, imputing would introduce inaccurate data
Duplicate records	None found — no action needed
Date format (ORDERDATE)	Already native DATE type — no action needed
Text casing (customer, city, contact names)	Corrected using INITCAP for consistent title case
Category fields (status, product line, etc.)	Confirmed already consistent — no action needed
PHONE formatting	Stripped to digits + plus sign only, via REGEXP_REPLACE
Invalid PRICEEACH (46% flat placeholder)	Added PRICEEACH_CORRECTED column, calculated as $\text{SALES} \div \text{QUANTITYORDERED}$. Original retained for auditability
Statistical outliers (SALES, QUANTITYORDERED)	Reviewed via IQR method, retained — legitimate large wholesale orders, not errors

\$10.03M in total revenue, from 307 orders

TOTAL REVENUE

\$10.03M

307 ORDERS · \$3,553.89 AVG ORDER LINE

2003	\$3.52M
2004	\$4.72M
2005*	\$1.79M

*Jan-May only, partial year

Is the business growing?

Revenue grew from \$3.52M (2003) to \$4.72M (2004), a 34% increase. 2005 data covers January through May only — annualized at the same pace, it tracks to roughly \$4.30M, a modest 9% pace decline versus 2004.

BUSINESS ANSWER

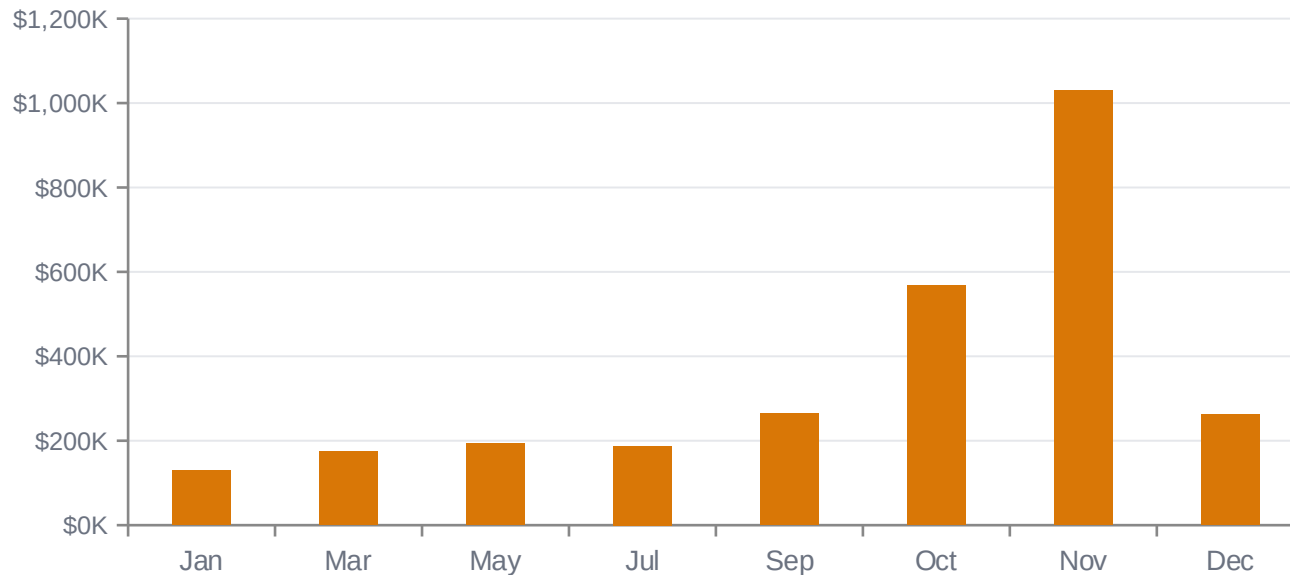
That 9% figure should be read with caution — it's based on five months of actual data, annualized. The honest framing is “insufficient data to confirm a declining trend,” not a definitive finding.

ORDER FULFILLMENT HEALTH

92.7% of order lines (\$9.29M of \$10.03M) are cleanly Shipped. Cancelled orders account for \$194,487 (1.9%) across 60 lines — small, but worth monitoring. Disputed orders are the smallest category (\$72K) but often signal service or quality issues worth a closer look.

November revenue runs 3-4x above a typical month

2003 Monthly Revenue — Seasonal Pattern



BUSINESS ANSWER

November revenue: \$1.03M (2003), \$1.09M (2004) — 3-4x a typical month, both years. This is backed by real order volume, not a few big orders: 28 distinct orders in Nov 2003, 33 in Nov 2004, against a ~9-12 order monthly average. October shows a smaller pre-peak surge.

Growth rates confirm it: Oct 2003 +115%, Nov 2003 +81%. Oct 2004 +72%, Nov 2004 +97%. Both years then crash: Dec 2003 -75%, Dec 2004 -66%.

Combined with the product mix (motorcycles, classic cars, collectibles), this points to holiday-season stocking by wholesale customers ahead of December retail demand.

RECOMMENDATION

Align inventory planning, staffing, and cash flow forecasting around the Oct-Nov surge. Consider early-bird incentives to shift some volume earlier in Q3/Q4 for a smoother operational load.

Q4 is the strongest quarter in every year on record



BUSINESS ANSWER

2003 Q4: \$1.86M — more than 4x Q1's \$445K.

2004 Q4: \$2.01M — nearly 2.5x Q1's \$834K.

Q4 is the strongest quarter in both complete years, Q1 consistently the weakest. This is the quarterly confirmation of the November stocking-surge finding from the previous slide — same pattern, wider lens.

One product, one category, well ahead of everything else

TOP SELLING PRODUCT

S18_3232

Classic Cars · \$288,245 revenue

Nearly 50% more revenue than the #2 product

7 OF 10

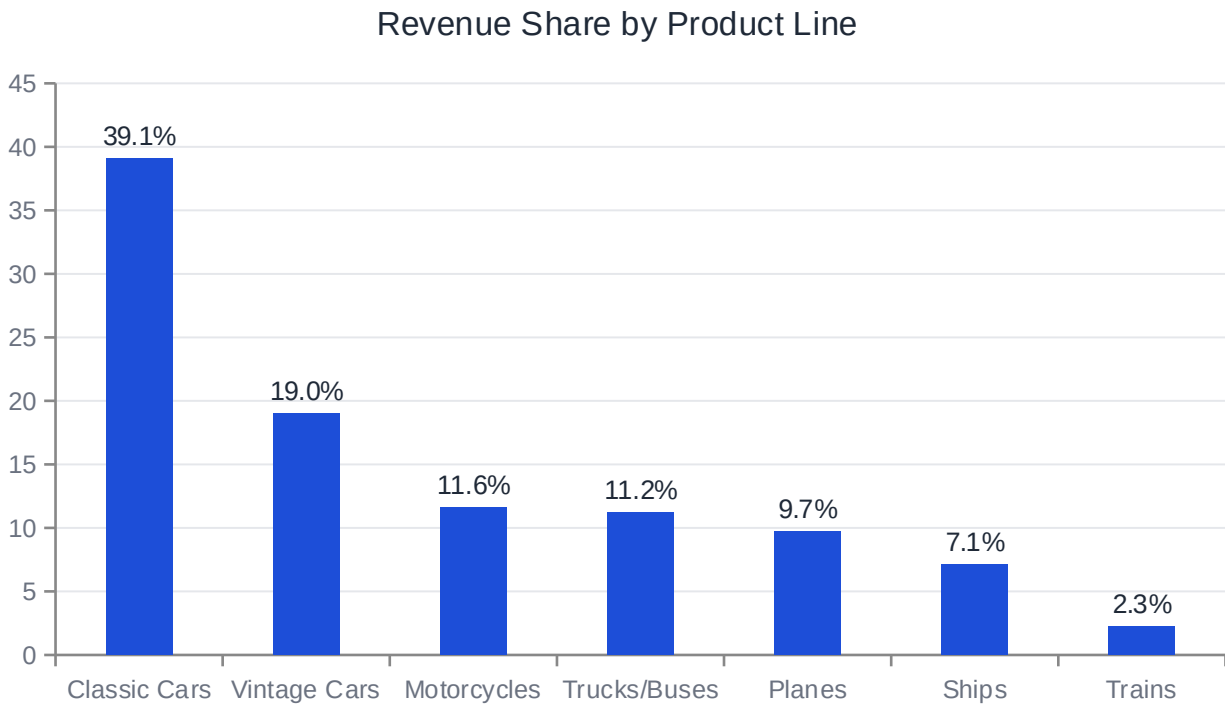
of the top 10 best-selling products by revenue are Classic Cars — category dominance at the individual SKU level, not a diversified catalog spread.

BUSINESS ANSWER & RECOMMENDATION

The single best-selling product, S18_3232, generated \$288,245 — nearly 50% more than the next closest product. More significant than any one SKU though: 7 of the top 10 products by revenue are Classic Cars, indicating this category is the primary revenue driver at the SKU level.

Recommend prioritizing inventory availability and marketing focus on Classic Cars, and specifically safeguarding supply and demand planning around S18_3232 given its outsized individual contribution.

Classic Cars alone drives 39% of total revenue



BUSINESS ANSWER

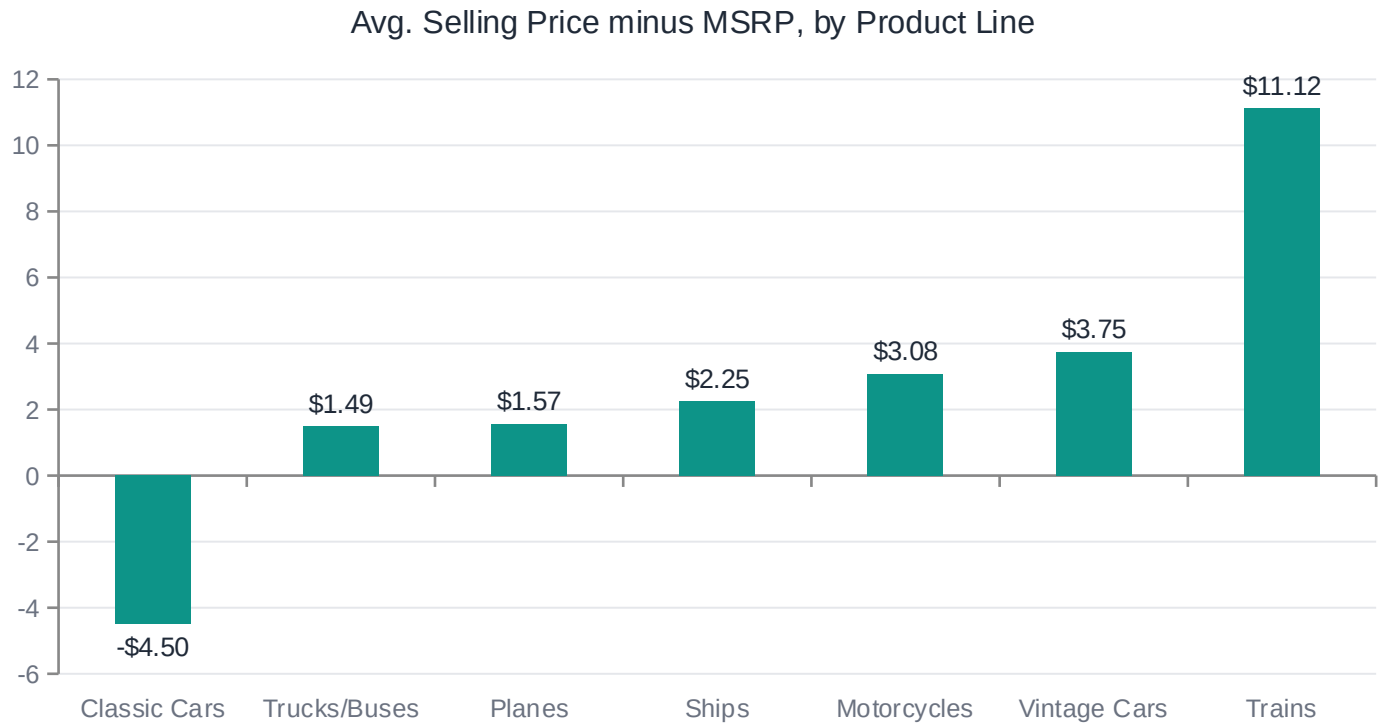
Classic Cars: 39.07% of total revenue from 37 products — more than double Vintage Cars (18.97%). Together, both car lines account for 58% of total revenue: significant category concentration risk.

Trains is smallest by far — 2.26% from only 3 SKUs — worth a review on whether it remains strategically worthwhile.

Separately: Medium-sized deals dominate revenue in every single product line without exception — a structurally healthy pattern, not dependent on a few large, high-risk deals. Ships shows no Large deals at all.

RECOMMENDATION: Prioritize supply chain resilience for Classic/Vintage Cars given the concentration risk. Consider whether growing Ships, Trucks and Buses, or Planes is worth pursuing to diversify.

Most product lines sell above their listed MSRP



BUSINESS ANSWER

Using the corrected unit price ($\text{SALES} \div \text{QUANTITYORDERED}$), most lines sell above MSRP — most notably Trains (+\$11.12) and Vintage Cars (+\$3.75). Classic Cars is the one exception, selling slightly below MSRP (−\$4.50), possibly reflecting volume discounting on the best-selling category.

Selling above MSRP is unusual for typical retail — may mean MSRP here is a wholesale baseline, not a retail list price. Worth clarifying against the original data source if possible.

One customer accounts for 9.1% of all revenue

TOP CUSTOMER

Euro Shopping Channel

Spain

\$912,294

revenue · 26 orders · ~9.1% of total company revenue

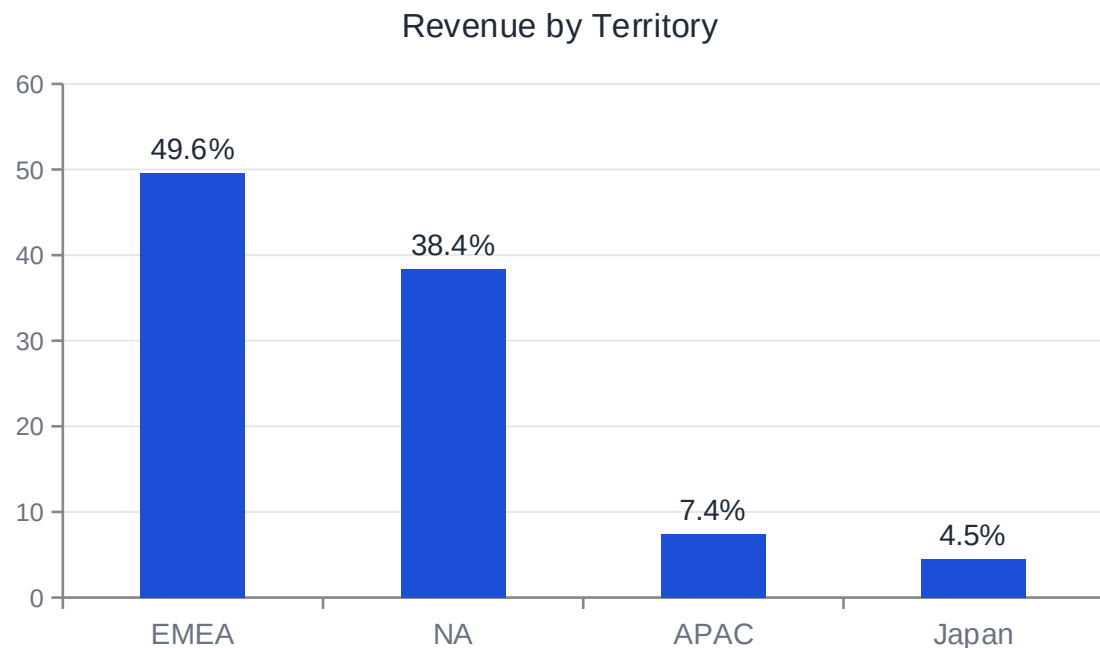
BUSINESS ANSWER

This is a meaningful customer concentration risk — losing this single account would materially impact overall revenue. Recommend prioritizing relationship management and retention strategy specifically for this account.

CUSTOMER CONCENTRATION — TOP 10

The top 10 customers (10.9% of the 92-customer base) generate 29.45% of total revenue (\$2.95M) — moderate concentration, not extreme, but significant enough to warrant active account management across the full top-10 group, not just the #1 account.

EMEA leads on breadth. The USA leads on concentration.



BUSINESS ANSWER

EMEA and North America drive 88% of revenue combined. EMEA leads (49.63%) but through breadth — 44 customers spread across Spain, France, UK, Italy and more, no single dominant country.

North America's share is driven almost entirely by one country: the USA alone generates \$3.63M — 36% of total company revenue, more than 3x the next country (Spain, \$1.22M).

So the USA, not EMEA, is the company's single greatest country-level concentration risk — even though EMEA leads at the territory level.

RECOMMENDATION

Monitor USA-specific market risk given its concentration. APAC and Japan are just 12% of revenue combined — worth assessing whether targeted growth investment there could reduce long-term dependency on the USA and EMEA.

KEY TAKEAWAYS

What I'd tell the business owner

1

Revenue is healthy but seasonally lopsided

\$10.03M total, growing 34% from 2003 to 2004 — but Oct–Nov drives a disproportionate share every year. Plan inventory and cash flow around it.

2

One product category is doing most of the work

Classic Cars alone is 39% of revenue; combined with Vintage Cars, 58%. Real concentration risk worth actively managing.

3

Revenue is built on volume, not big bets

Medium-sized deals dominate every product line — a structurally healthy pattern, not dependent on a few large, risky transactions.

4

Two accounts to watch closely

Euro Shopping Channel (9.1% of revenue) and the USA as a country (36% of revenue) are the two biggest single points of dependency.

5

Fulfillment is solid

92.7% of order lines are cleanly shipped. Cancellations and disputes are small but worth monitoring as leading indicators.

Thank you

Full queries, audit findings, cleaning process, and business insights are documented in the project repository.

PostgreSQL · pgAdmin · SQL (joins, CTEs, window functions, aggregates, IQR outlier detection)

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